

Vaibhav Lalwani

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SUMMARY

AI engineer building production language-model systems end-to-end — inference, retrieval, evaluation harnesses and the front ends that ship them. MSc Advanced Data Science & AI at the University of Liverpool. Author of an open-access preprint on on-controller transformer inference (NEXUS, 2026). Comfortable across the model–system boundary.

PUBLICATIONS

Lalwani, V. (2026). *NEXUS: On-Controller Transformer Inference and Speculative Edge Execution for Console-Wired Latency Parity in Cloud Gaming*. Zenodo. DOI: [10.5281/zenodo.20059414](https://doi.org/10.5281/zenodo.20059414) · ResearchGate. Extends the Outatime line (MobiSys 2015) with on-controller transformer inference and speculative edge execution.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript (ES6+), SQL, HTML, CSS

AI / ML: OpenAI, Anthropic / Claude, Gemini, Groq, LangChain, LangSmith, RAG pipelines, prompt engineering, structured outputs, MCP servers, Whisper STT, PyTorch (familiar), transformer inference (NEXUS paper)

Backend & Data: FastAPI, Node.js / Express, Postgres, pgvector, Redis, BullMQ, Supabase (RLS), vector search

Frontend: React 19, Next.js 15 / 16, Tailwind CSS, Vite, REST + GraphQL, accessibility basics

DevOps: Docker, Git, GitHub Actions, Vercel, GCP Cloud Run, AWS, Sentry, PostHog, Jest / Vitest, Playwright, Puppeteer

EXPERIENCE

Full-Stack / AI Engineer Intern

Apr 2026 – Present

Meta Solution Technologies

Remote, UK

- Building an AI-guided admissions platform on Next.js 15 + React 19 + TypeScript. Chat-style guidance flow runs on OpenAI with structured outputs against a Postgres + pgvector knowledge store, served from a Python FastAPI service.
- Wrote auth end-to-end (Google OAuth, magic-link, PKCE) on Supabase, plus a BullMQ + Redis async queue driving document checks, generation jobs and email triggers.
- Shipped a 10-language i18n layer and a dual-theme CSS system so non-engineering teammates roll out copy and translation updates without involving me.

AI Engineer (Freelance)

May 2024 – Jan 2026

Self-Employed

Remote

- Designed and shipped 20+ LLM-powered applications for paying clients across SaaS, e-commerce and consulting — most pairing a React or Next.js front end with a Python service running Claude or OpenAI APIs, structured outputs and a RAG retrieval layer.
- Built longer-running AI agents for lead generation, customer support and data processing. Each project shipped with a README, a walkthrough video and a written hand-off doc.

Software Engineer Intern

Jan 2026 – Apr 2026

Recruit Pilot — Recruitment Technology

Remote

- Built React + TypeScript UI features against a REST API for a recruitment-tech product. Code review and weekly sprint cadence alongside more senior engineers.

SELECTED PROJECTS

apex — autonomous job application engine. CLI that drives real Chrome via Playwright, generates a tailored 1-page resume per job (Puppeteer markdown → ATS-safe PDF), fills custom LinkedIn Easy Apply questions with an LLM (profile mapping → Q+A cache → free-LLM fallback) and submits autonomously until LinkedIn's daily cap. Free LLMs only; pause-on-stuck UX hands off when a required field can't be answered confidently. github.com/vaibhav4046/apex

Praxon — open-source AI agent platform. A Claude Cowork alternative built end-to-end on Next.js 16 + React 19 + TypeScript. Multi-LLM router across free providers (Groq, Cerebras, Gemini, Ollama) with auto-fallback; MCP-native tool layer; 3-schema Postgres design with RLS for tenant isolation; cloud-deployable on Vercel + Supabase. [Live](#) · [Code](#)

Cogniloop — Socratic study tool. Locked-prompt evaluator that grades free-form student answers against a rubric, scores 0–3 with explanation, tracks concept mastery (weak → mastered) across sessions. Single-prompt design — no agent loop, no retrieval — keeps median latency under one second on Groq Llama 3.3 70B with a Pollinations fallback. Edge runtime, Next.js 16, KaTeX. [Live](#) · [Code](#)

MCP Marketplace — registry of 800+ Model Context Protocol servers. Daily auto-sync from Glama and the official MCP repo, normalised tool schemas, one-line install snippets for Claude Desktop, Cursor and Claude Code. Next.js 15 RSC, Cmd-K palette, dynamic OG cards, full keyboard navigation. [Live](#) · [Code](#)

EDUCATION

University of Liverpool

MSc Advanced Data Science & Artificial Intelligence

Jan 2026 – Jan 2027

Christ University, Bengaluru

Bachelor of Computer Applications · CGPA 7.86 / 10

2022 – 2025